

BLUESTOCK

FINTECH

(bluestock.in)

MUTUAL FUND ANALYTICS PLATFORM

FINAL INTERNSHIP REPORT

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Submission Year	2026

Submitted to BlueStock Fintech

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2 Executive Summary

MUTUAL FUND ANALYTICS PLATFORM

Project Workflow — Executive Summary

01

CONTEXT

Built a centralized, interactive Mutual Fund Analytics Platform at BlueStock Fintech to address database fragmentation in retail wealth management.

02

DATABASE

Structured a star-schema SQLite database integrating ten disparate CSV datasets — funds, historical NAVs, state-wise transactions, and stock portfolios.

03

ETL PIPELINE

Programmed a Python-based ETL pipeline performing automated data cleanups, key mapping fixes, and daily API-driven valuation updates.

04

VISUAL INTERFACE

Created a 12-page analytical dashboard in Streamlit displaying performance tables, state-wise transaction charts, and SIP trends.

05

ADVANCED ANALYTICS

Integrated portfolio optimization (Markowitz Mean-Variance Efficient Frontier) and future asset growth simulators (Monte Carlo paths).

06

BUSINESS OUTCOME

Delivered direct PDF/Excel analytical download modules and an administrative log desk, significantly reducing manual reporting overhead for wealth advisors.

Live link-<https://bluestock-mf-analytics.streamlit.app/>

Note: I scheduled my local ETL pipeline to run automatically every day at 8:00 PM on my computer. However, this only updates the database file locally on my machine. Since the live website reads the database file hosted on my GitHub repository (and Streamlit Cloud servers reset periodically), I must push the updated local database to GitHub to show the daily 8:00 PM updates on the live website. I have to push it manually every time because my local computer is isolated from the cloud, and the Streamlit Cloud server has no direct access to read files from my local system.

3 Problem Statement & Objectives

3.1 Problem Statement

- **Data Fragmentation:** Mutual fund details, daily historical NAVs, benchmark values, and sector allocations are published by different entities in diverse formats, preventing holistic analysis.
- **Lack of Risk-Adjusted Tools:** Retail investors often judge funds solely on past returns (e.g., 1Y or 3Y returns) without evaluating volatility, drawdowns, or risk-adjusted return ratios (Sharpe, Sortino).
- **Static Advisory Reports:** Traditional wealth management utilizes static PDF reports that fail to allow interactive screening, portfolio reweighting, or custom horizon projections.
- **Manual Pipelines:** Wealth teams spend significant effort manually loading spreadsheet data, introducing errors.
- **Data Inconsistency:** Inconsistent scheme names and incorrect AMFI identifiers across sources lead to broken joins.

3.2 Project Objectives

- **Relational Consolidation:** Host all historical mutual fund records in a local, optimized database.
- **Automated Updates:** Set up a scheduled ETL pipeline that cleans data and pulls daily valuations incrementally.
- **Multi-Criteria Recommender:** Create a scoring engine to rank funds based on investor goals, risk, and expense limits.
- **Risk Metrics Computation:** Build calculators to compute Sharpe/Sortino ratios, Alpha, Beta, and Max Drawdowns.
- **Optimization Integration:** Implement mathematical asset weighting models using Markowitz Modern Portfolio Theory.
- **Report Generation:** Provide download support for client-ready PDF reports and structured Excel files.

4 Technology Stack

Component	Technology	Purpose	Usage in Project
Language	Python	Core logic and mathematical operations	Processes files, computes metrics, and controls app flow.
Dashboard	Streamlit	Multi-page web dashboard framework	Renders navigation tabs, KPIs, and visual charts.
Database	SQLite	Relational database engine	Hosts staging tables, dimensions, facts, and audit metadata.
Data Cleaning	Pandas	Structured data manipulation	Cleans, filters, joins, and aggregates relational tables.
Mathematics	NumPy / SciPy	Array math and portfolio optimization	Runs Monte Carlo scenarios and quadratic optimization.
Charting	Plotly	Interactive charting library	Renders line charts, bars, donuts, and heatmap grids.
Exports	ReportLab / open-pyxl	PDF fact sheets and Excel books	Generates formatted reporting files for advisors.

5 Database Design & ETL Pipeline

5.1 Database Schema

- **dim fund:** Columns: fund id (PK), amfi-code, fund-house, scheme name, category, sub-category, plan, risk category, benchmark, fund manager. Stores fund static details.
-
- **dim date:** Columns: date id (PK), full date, day, month, year, quarter, day of week. Maps timeline attributes for rapid time-series analysis.
-
- **fact nav:** Columns: nav id (PK), fund id (FK), date id (FK), nav. Stores daily valuation prices.
-
- **fact performance:** Columns: perf id (PK), fund id (FK), return 1yr pct, return 3yr pct, return 5yr pct, sharpe ratio, sortino ratio, alpha, beta, std dev ann pct, max drawdown pct, aum crore, expense ratio pct. Stores pre-calculated risk-return ratios.
-
- **etl metadata:** Logs date/time stamps, status, and row-counts of daily updates.

5.2 ETL Workflow and Data Cleaning

- **Extraction:** Reads raw CSV files and queries the AMFI / MFAPI APIs dynamically for live historical NAV data.
-
- **Transformation:** Standardizes calendar date fields, coerces strings to float values, removes duplicate indices, and handles missing values.
-
- **Identifier Correction:** Corrects AMFI identification mismatches (ABSL Large Cap from 112114 to 103174; DSP Top 100 from 105872 to 101635) to match official registries.
-
- **Loading:** Loads cleaned staging tables into SQLite, enforcing foreign keys and unique constraints.
-
- **Daily Incremental Update:** Checks the database for the last-logged date, fetches new daily NAV data since that date, appends it to fact nav, and audits the run in etl metadata.

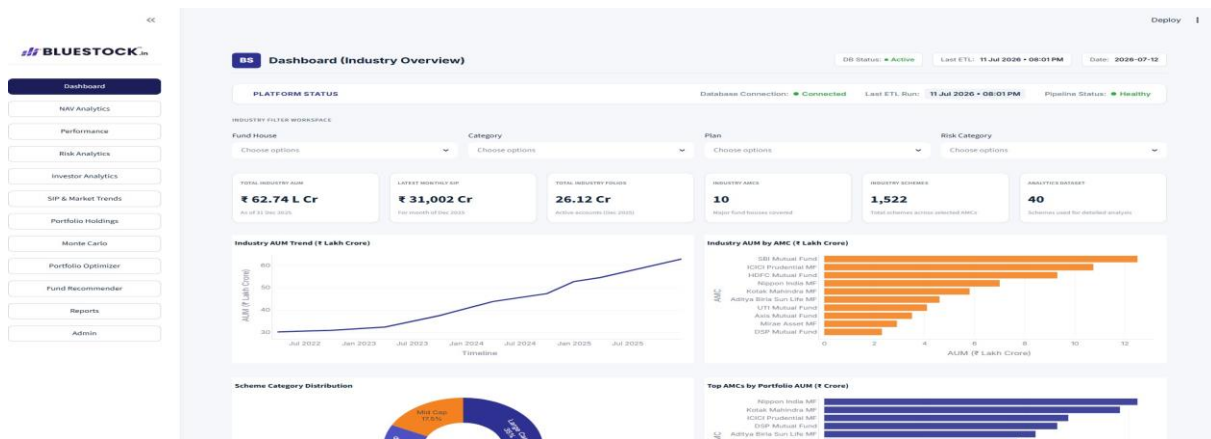
6 Dashboard Overview & Page Details

6.1 Dashboard Layout

- **Framework:** Streamlit Multi-Page structure with sidebar navigation.
- **Page Organization:** A 12-page structure mapping distinct analytical areas.
- **Status Panel:** Displays database active status and the timestamp of the last ETL run on the sidebar.

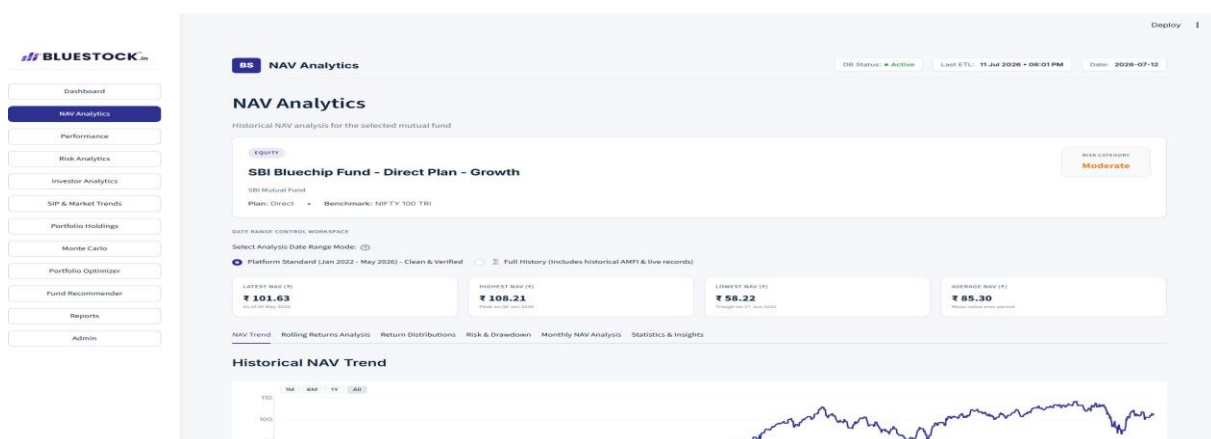
6.2 Page-wise Features

Page 1: Dashboard (Overview)



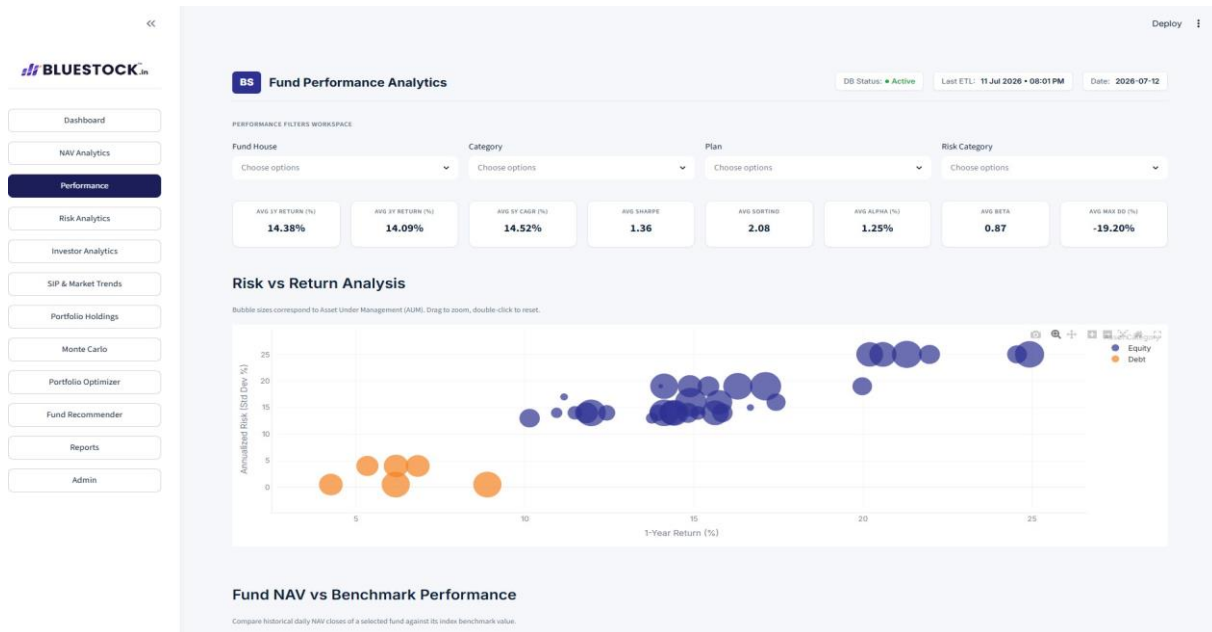
- **Purpose:** Summarizes overall mutual fund assets, AMC market shares, and fund categories.
- **KPIs:** Total Industry AUM, Active Fund Houses, and Category Splits.
- **Charts:** Industry AUM Trend, AMC AUM rankings, and category allocation pie.

Page 2: NAV Analytics



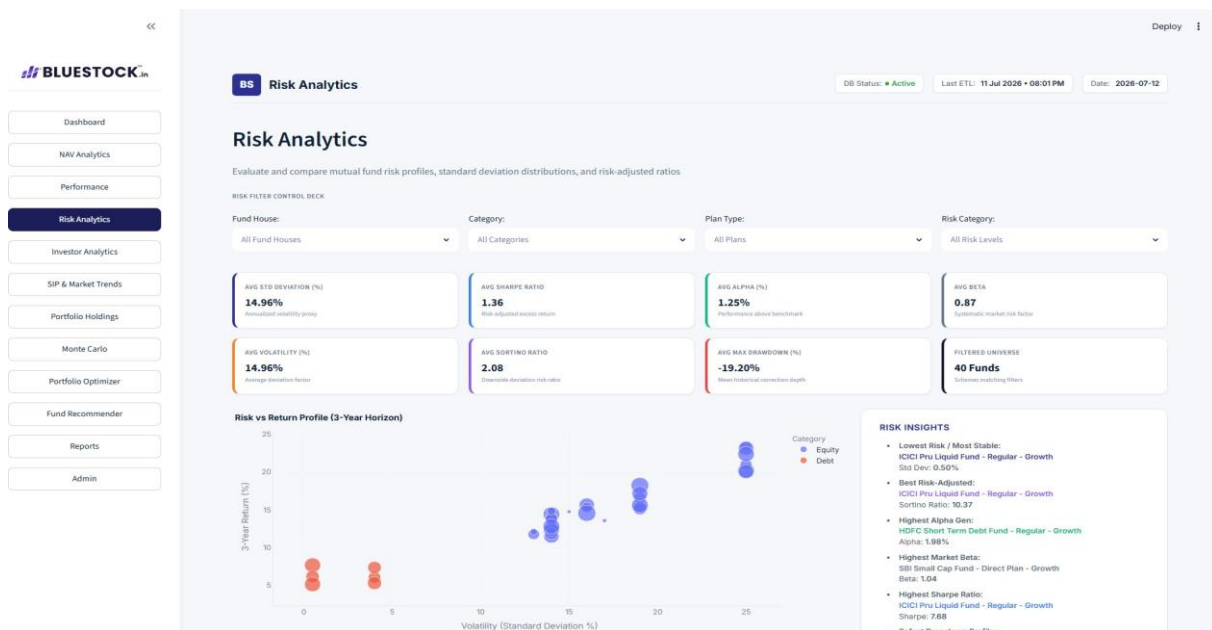
- **Purpose:** Detailed daily evaluation for a selected scheme.
- **Features:** Period selectors (1M, 6M, 1Y, All), rolling returns, indicator tables, drawdowns, and calendar return heatmaps.

Page 3: Performance



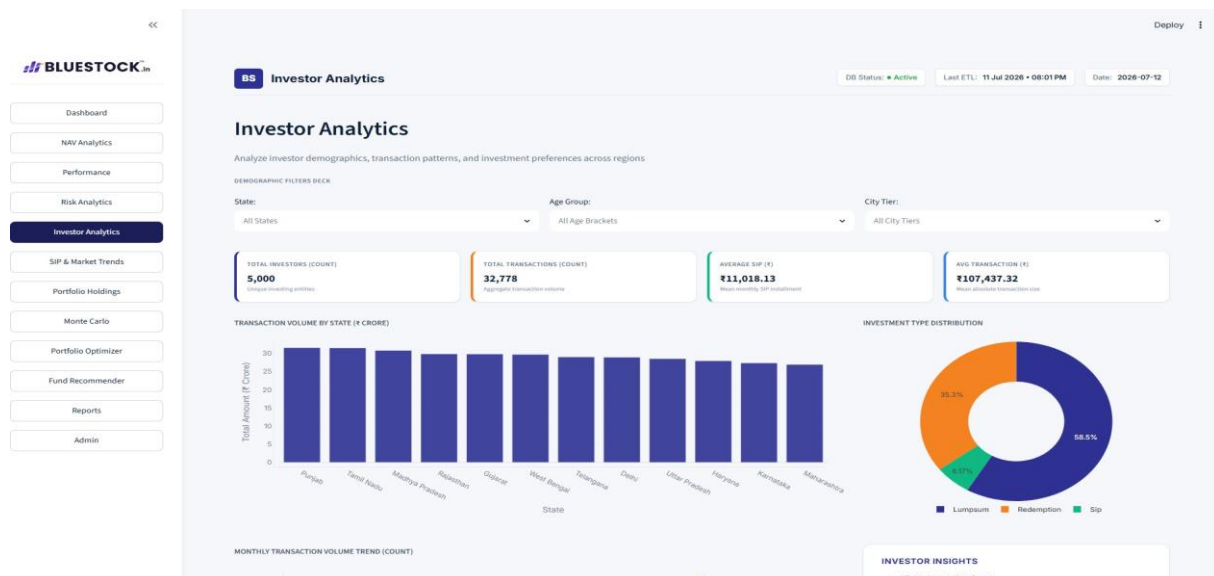
- **Purpose:** Compares risk, returns, and benchmark indexes.
- **KPIs:** Mean returns, benchmark comparisons, volatility limits, and alpha metrics.
- **Features:** Returns distribution histograms and fund-benchmark dual-axis line charts.

Page 4: Risk Analytics



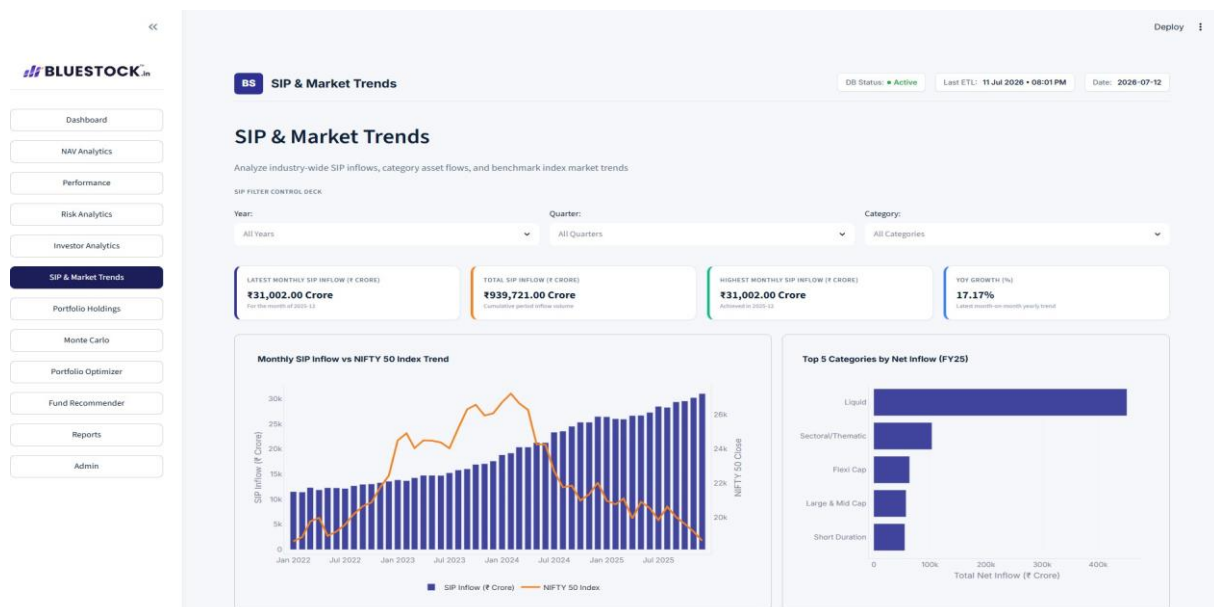
- **Purpose:** Dedicated to identifying fund risk characteristics and volatility levels.
- **KPIs:** Standard deviation bounds, system beta values, and Sharpe/Sortino values.
- **Features:** Volatility vs return scatter, volatility distribution, and Sharpe ratio comparisons.

Page 5: Investor Analytics



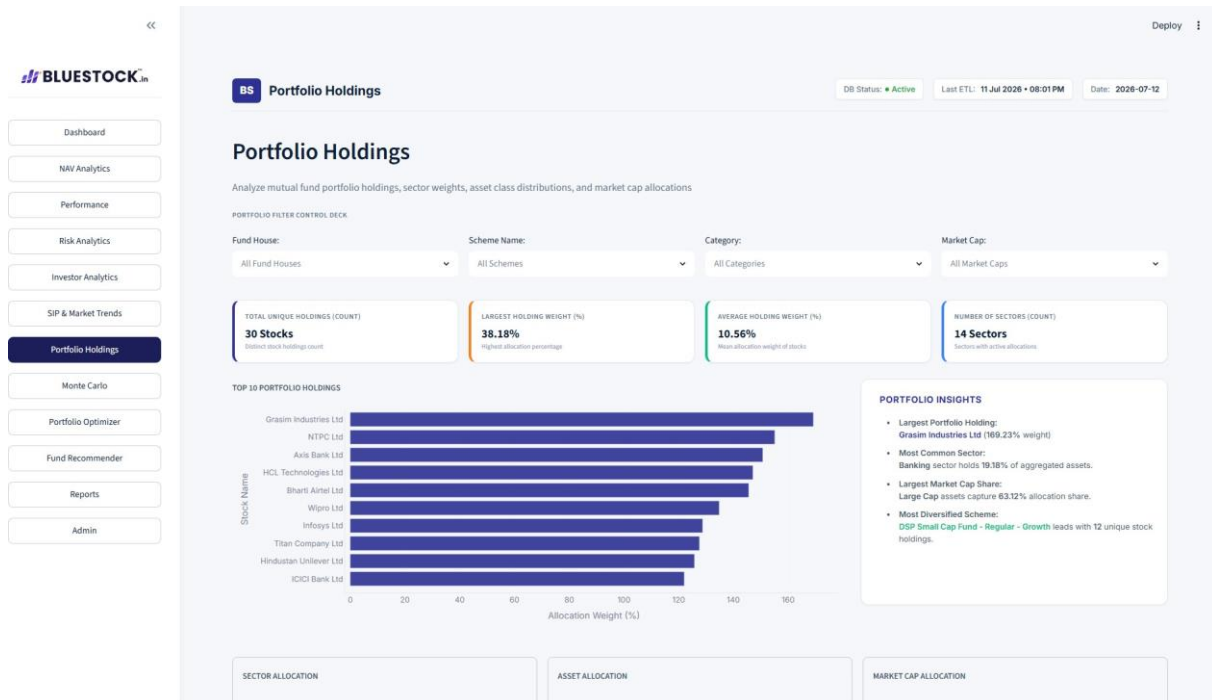
- **Purpose:** Tracks investor demographics, locations, and saving channels.
- **KPIs:** Total Investors, Transaction volume, and Average transaction size.
- **Features:** Transaction volume by state, investment type splits, monthly count trends, and age demographics.

Page 6: SIP & Market Trends



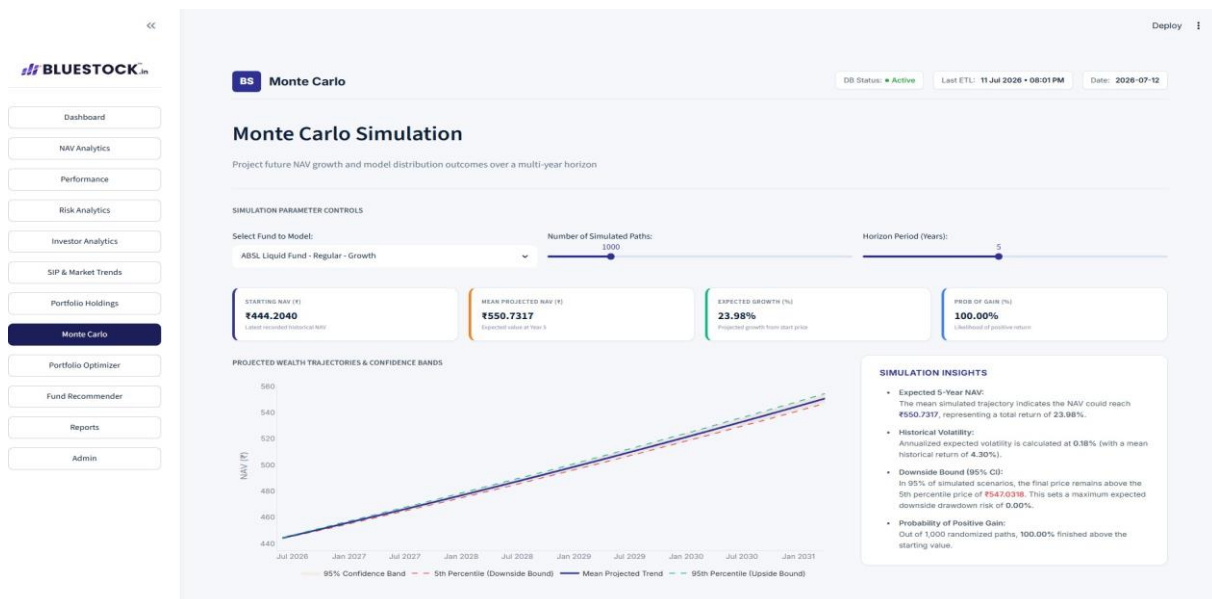
- **Purpose:** Compares retail systematic saving patterns against broader market indices.
- **KPIs:** Monthly inflow growth rate, peak monthly collections, and folio growth.
- **Charts:** Monthly SIP vs Nifty 50 close, Category-wise net inflow, and inflow heatmap.

Page 7: Portfolio Holdings



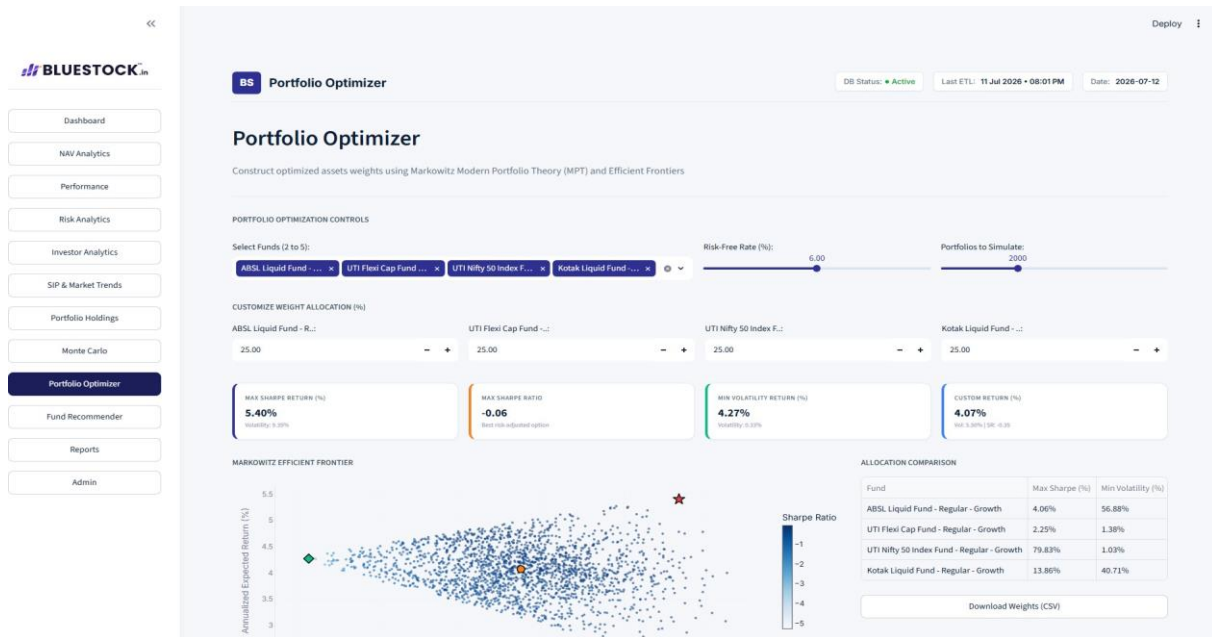
- **Purpose:** Analyzes portfolio stock allocations and concentration risks.
- **KPIs:** Unique Holdings Count, Largest stock weight, and Number of active sectors.
- **Charts:** Top 10 holdings bar chart, Sector allocation pie, Asset class pie, and Market cap distribution.

Page 8: Monte Carlo Simulation



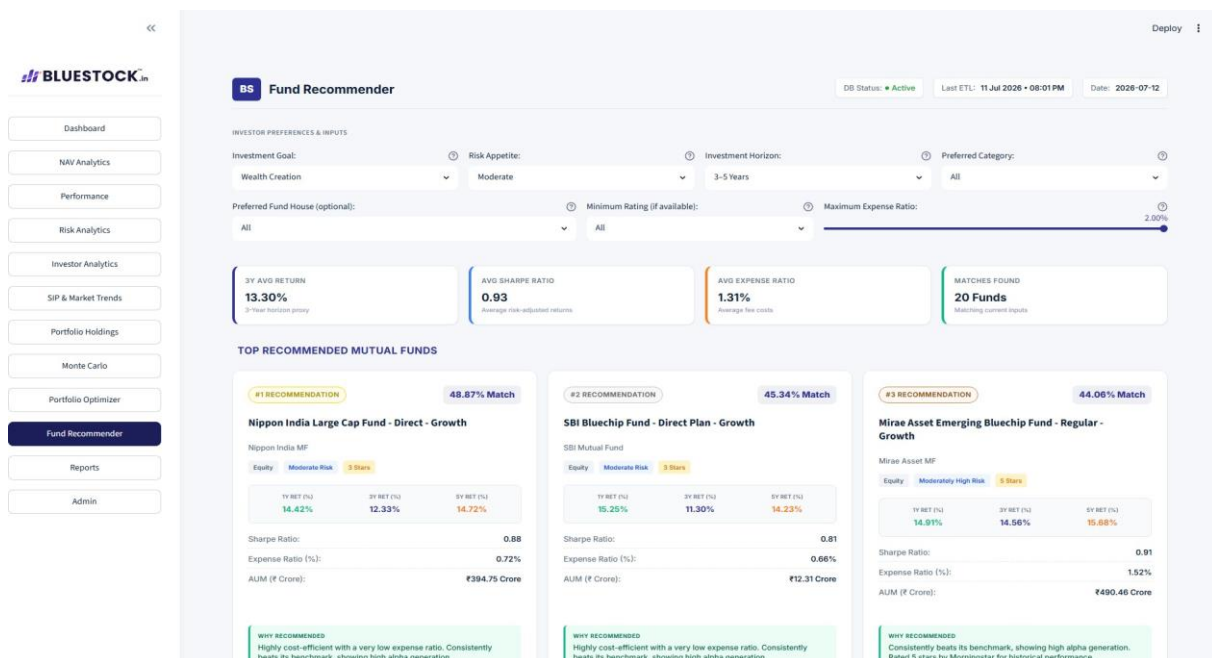
- **Purpose:** Projects future value distributions of portfolio assets using probability simulations.
- **Interactive Inputs:** Principal investment (Rs.), target horizon (years), return projection %, and scenario count.

Page 9: Portfolio Optimizer



- **Purpose:** Computes asset weights to maximize risk-adjusted efficiency.
- **Visualizations:** Portfolio Efficient Frontier scatter plot, highlighting Max Sharpe and Min Volatility portfolios.

Page 10: Fund Recommender



- **Purpose:** Suggests mutual funds based on client risk levels and targets.
- **Inputs:** Investment Goal, Risk Appetite, Horizon, and Expense Ratio limits.
- **Outputs:** Match percentage scores, Top 3 recommended cards, qualitative reasons, and candidate grids.

Page 11: Reports & Export

BS Reports DB Status: Active Last ETL: 11 Jul 2026 • 08:01 PM Date: 2026-07-12

Reports & Export

Download custom tabular CSV datasheets, styled Excel sheets, and executive PDF briefing reports

Choose Report Dataset to Export:
NAV Analytics History

Search in report data:
Type keywords (e.g. Nippon, Equity, Karnataka)...

Filter by Category:
All

PREVIEWING TOP 10 OF 1,000 MATCHING RECORDS

Scheme Name	Category	Date	NAV (₹)
Kotak Liquid Fund - Regular - Growth	Debt	2026-07-11 00:00:00.000000	5617.9669
ABSL Liquid Fund - Regular - Growth	Debt	2026-07-10 00:00:00.000000	448.0221
UTI Flexi Cap Fund - Regular - Growth	Equity	2026-07-10 00:00:00.000000	316.4588
UTI Nifty 50 Index Fund - Regular - Growth	Equity	2026-07-10 00:00:00.000000	167.5587
Kotak Liquid Fund - Regular - Growth	Debt	2026-07-10 00:00:00.000000	5617.0668
SBI Magnum Gilt Fund - Regular Plan - Growth	Debt	2026-07-10 00:00:00.000000	68.4996
DSP Top 100 Equity Fund - Regular - Growth	Equity	2026-07-10 00:00:00.000000	461.24
HDFC Top 100 Fund - Regular Plan - Growth	Equity	2026-07-10 00:00:00.000000	1136.976
UTI Mid Cap Fund - Regular - Growth	Equity	2026-07-10 00:00:00.000000	310.7476
ICICI Pru Midcap Fund - Regular - Growth	Equity	2026-07-10 00:00:00.000000	338.57

GENERATE AND DOWNLOAD FILES

Download CSV Sheet Download Excel Spreadsheet Download PDF Report

- **Purpose:** Generates reports for offline advisory meetings.
- **Downloads:** Direct client fact sheet PDFs, CSV datasets, and multi-sheet formatted Excel files.

Page 12: Admin Panel

BS Admin DB Status: Active Last ETL: 11 Jul 2026 • 08:01 PM Date: 2026-07-12

Administration Panel

Monitor database diagnostics, clear memory cache tables, and trigger automated HTML email briefings

System Diagnostics Automated HTML Emails (BS)

DATABASE SIZE (MB)
8.07 MB
OS file storage weight

TOTAL DATABASE TABLES (COUNT)
12 Tables
OS file storage weight

ETL HEALTH STATUS
Healthy
Last ETL: 11 Jul 2026 • 08:01 PM

LAST NAV DATE
2026-07-11
Last daily history entry

TABLE ROW COUNT SUMMARIES

Table Name	Row Count (Count)
dim_fund	40
dim_date	6,599
fact_nav	171,489
fact_transactions	32,778
fact_performance	40
stg_portfolio_holdings	322
stg_monthly_sip_inflows	48
stg_benchmark_indices	8,050

ADMINISTRATIVE ACTIONS

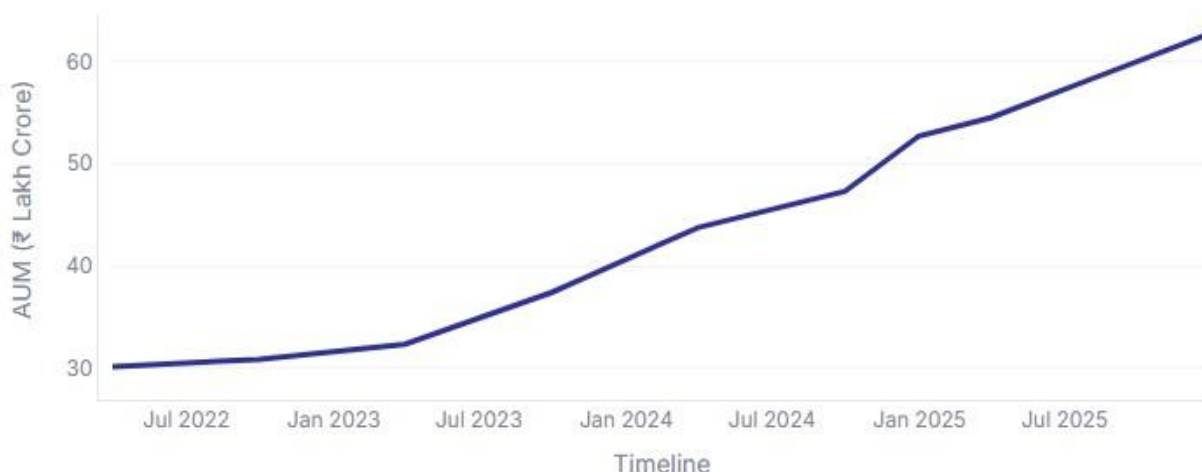
Trigger ETL Data Refresh Flush Memory Cache Tables Recompute Risk Metrics

- **Purpose:** System diagnostics dashboard showing ETL and database health.
- **KPIs:** Database active status, Last ETL run, total rows, and live logs.

7 Visualizations

7.1 Industry AUM Trend

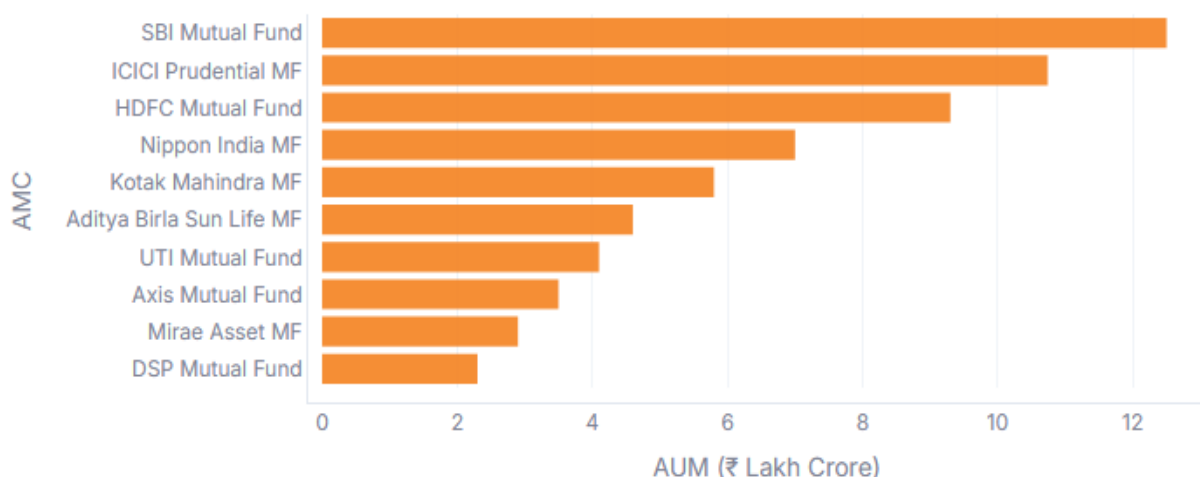
Industry AUM Trend (₹ Lakh Crore)



- **File Name:** industry_aum_trend.png | **Page Location:** Dashboard (Page 1)
- **Purpose:** Tracks mutual fund industry asset size growth.
- **X-Axis / Y-Axis:** Date Timeline (X) vs Industry AUM in Rs. Lakh Crore (Y).
- **Applied Filters:** Aggregated industry totals.
- **Business Insight:** Line chart illustrating growth of mutual fund industry assets, indicating increasing retail investor participation.

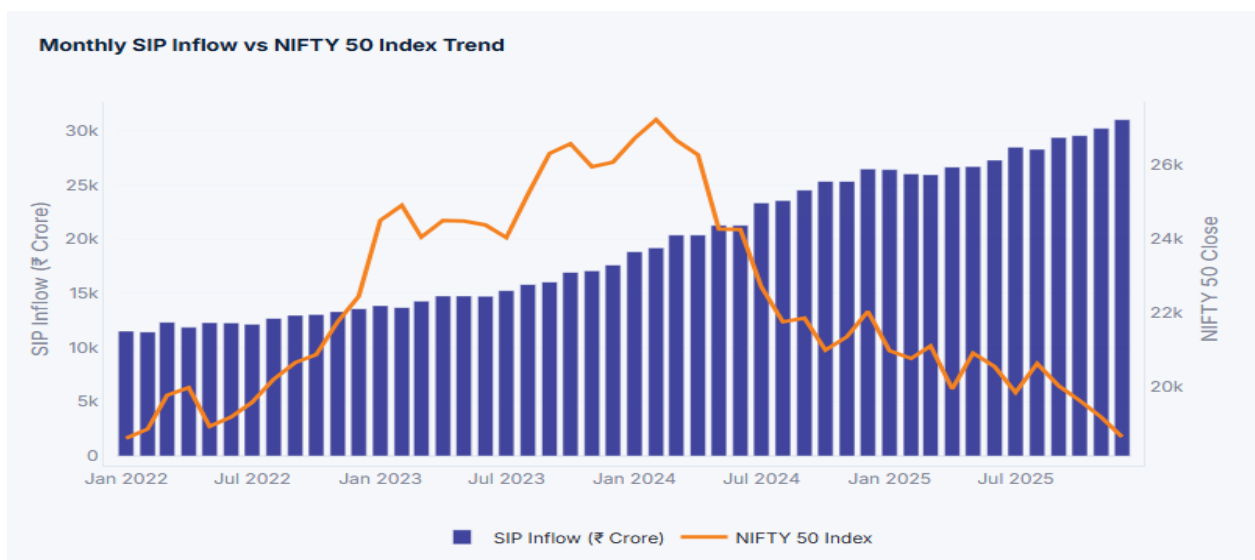
7.2 AUM by Fund House

Industry AUM by AMC (₹ Lakh Crore)



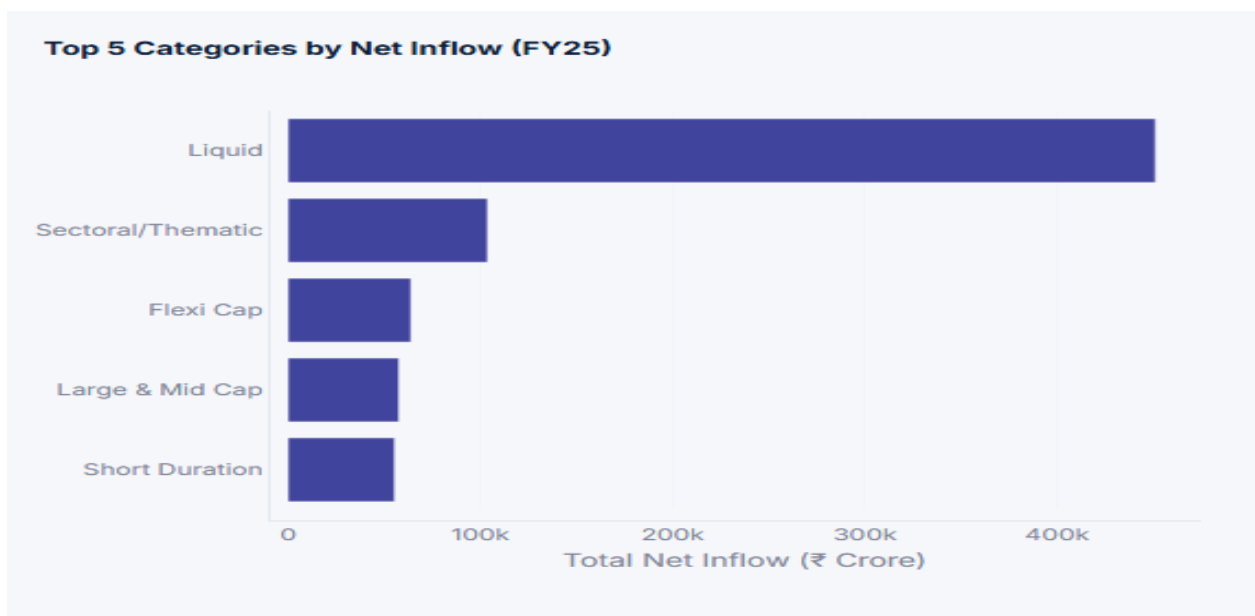
- **File Name:** aum_by_fund_house.png | **Page Location:** Dashboard (Page 1)
- **Purpose:** Compares asset sizes across Fund Houses (AMCs).
- **X-Axis / Y-Axis:** Asset Size in Rs. Lakh Crore (X) vs AMC Name (Y).
- **Applied Filters:** Sorted descending by latest AUM reports.
- **Business Insight:** Horizontal bar chart highlighting asset concentration, where a few top AMCs manage the majority of capital.

7.3 Monthly SIP Inflow Trend



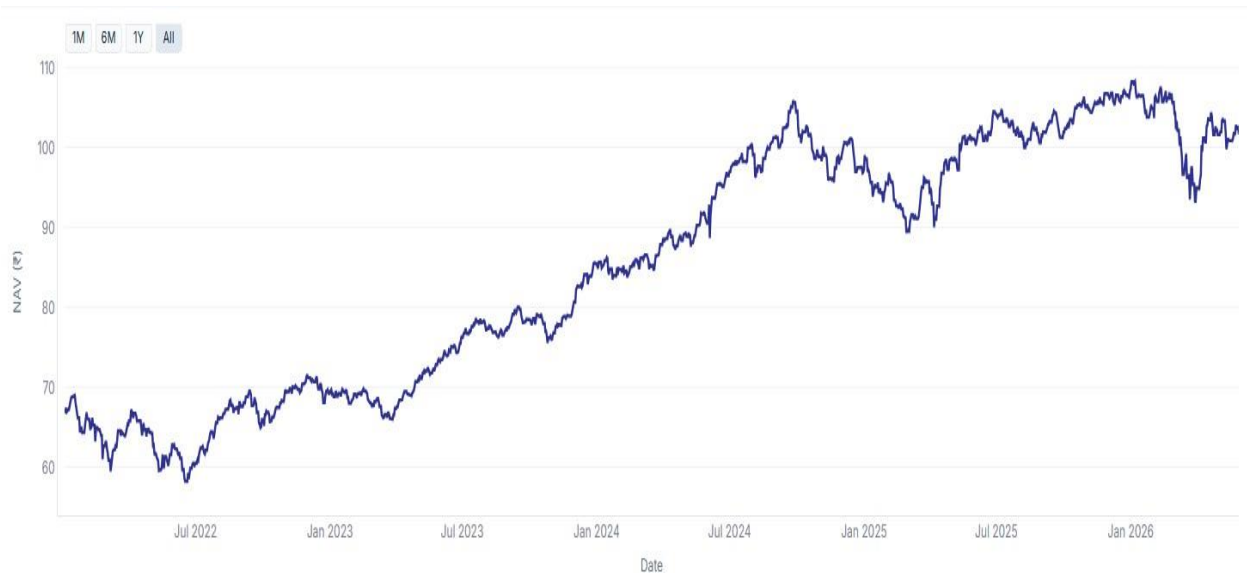
- **File Name:** monthly sip inflow trend.png | **Page Location:** SIP & Market Trends (Page 6)
- **Purpose:** Tracks retail saving patterns against market moves.
- **X-Axis / Y-Axis:** Month (X) vs Inflow in Rs. Crore [Bar] / Nifty 50 Close [Line] (Y).
- **Applied Filters:** Aggregated nationally across all categories.
- **Business Insight:** Combined chart showing that monthly SIP volumes remain stable even during market corrections.

7.4 Category-wise Inflow Analysis



- **File Name:** category wise inflow analysis.png | **Page Location:** SIP & Market Trends (Page 6)
- **Purpose:** Ranks fund categories based on recent inflows.
- **X-Axis / Y-Axis:** Total Net Inflow in Rs. Crore (X) vs Fund Category (Y).
- **Applied Filters:** Filtered for the latest FY25 reports.
- **Business Insight:** Bar chart showing which categories (e.g. Large Cap vs Mid Cap) are currently capturing the most capital.

7.5 Daily NAV Trend



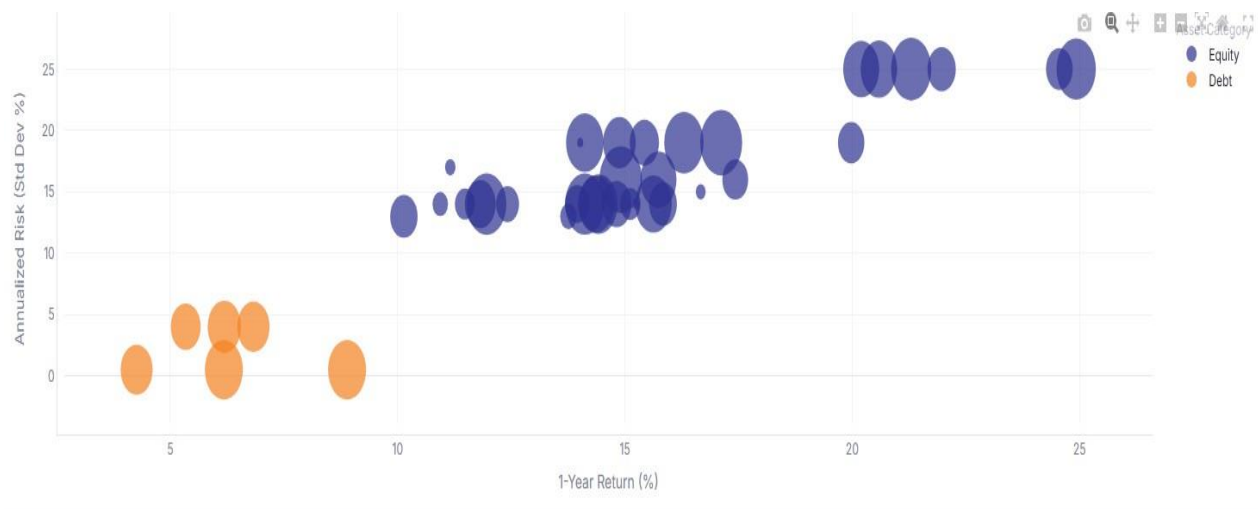
- **File Name:** daily nav trend.png | **Page Location:** NAV Analytics (Page 2)
- **Purpose:** Visualizes historical price performance for a selected scheme.
- **X-Axis / Y-Axis:** Timeline Date (X) vs Unit NAV in Rs. (Y).
- **Applied Filters:** Selected Scheme: *SBI Bluechip Fund – Direct Plan – Growth*.
- **Business Insight:** Displays compounding asset value and recovery durations for a specific fund.

7.6 Fund NAV vs Benchmark Performance



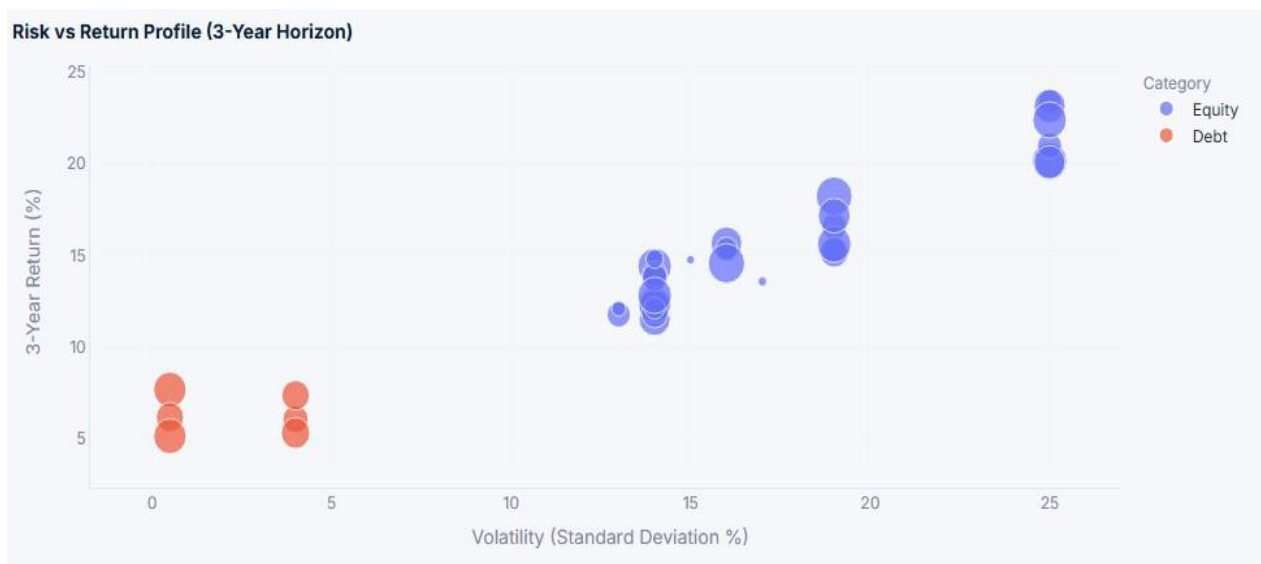
- **File Name:** fund nav vs benchmark.png | **Page Location:** Performance (Page 3)
- **Purpose:** Compares active fund returns directly against market benchmarks.
- **X-Axis / Y-Axis:** Date (X) vs Fund NAV in Rs. [Primary Y] and NIFTY100 index close [Secondary Y].
- **Applied Filters:** Selected Scheme: *ABSL Frontline Equity Fund – Regular – Growth* vs NIFTY100 Index.
- **Business Insight:** Displays whether the fund manager generated alpha over various market cycles.

7.7 Risk vs Return Bubble Chart



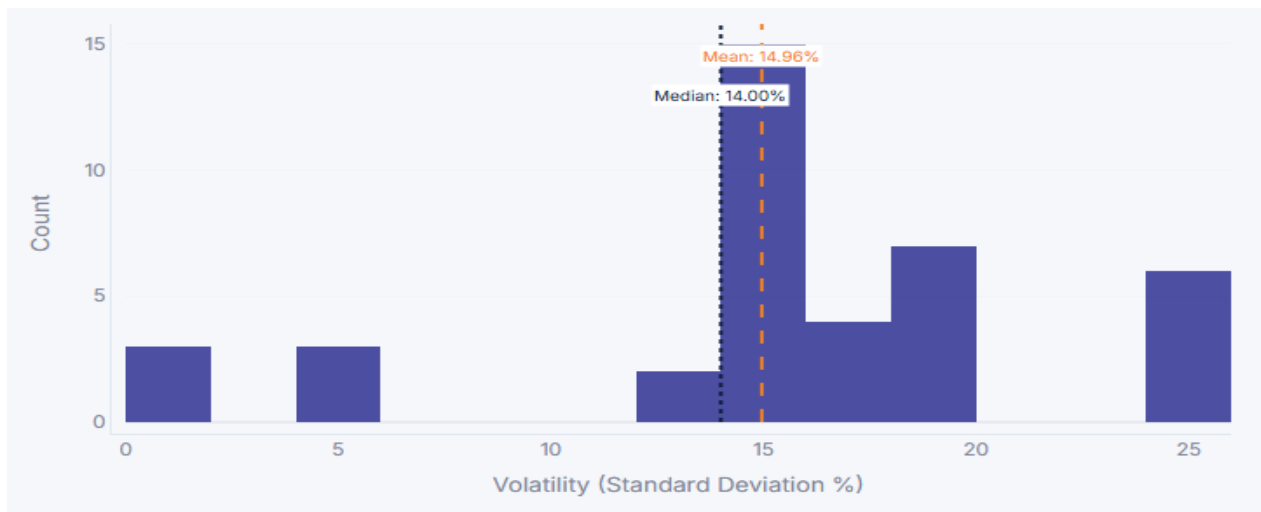
- **File Name:** risk vs return bubble.png | **Page Location:** Performance (Page 3)
- **Purpose:** Maps the risk-return characteristics of all funds.
- **X-Axis / Y-Axis:** 1-Year Return % (X) vs Volatility Standard Deviation % (Y).
- **Applied Filters:** Includes all active funds; bubble size represents AUM in Rs. Crore.
- **Business Insight:** Identifies efficient funds located in the upper-left quadrant (high return, low volatility).

7.8 Alpha vs Beta Scatter Plot



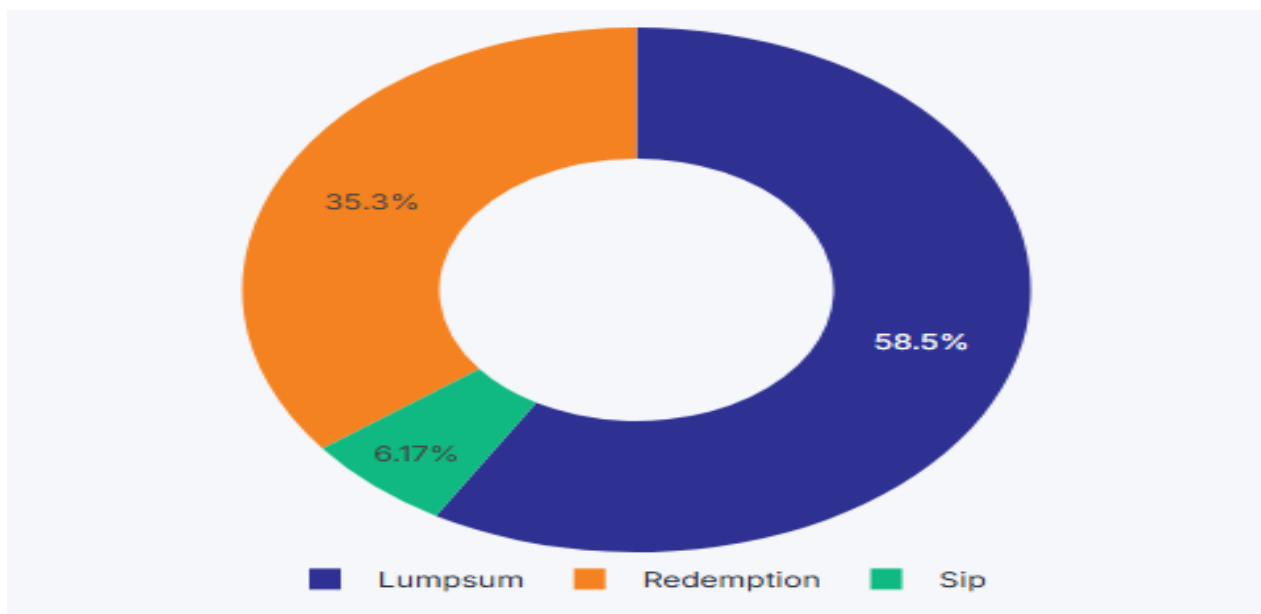
- **File Name:** alpha vs beta scatter.png | **Page Location:** Risk Analytics (Page 4)
- **Purpose:** Compares fund returns relative to volatility over a longer horizon.
- **X-Axis / Y-Axis:** Volatility Standard Deviation % (X) vs 3-Year Return % (Y).
- **Applied Filters:** Aggregated across the 3-year history; bubble size represents AUM in Rs. Crore.
- **Business Insight:** Illustrates that higher volatility does not always yield higher returns, highlighting the importance of risk-adjusted metrics.

7.9 Expense Ratio Comparison



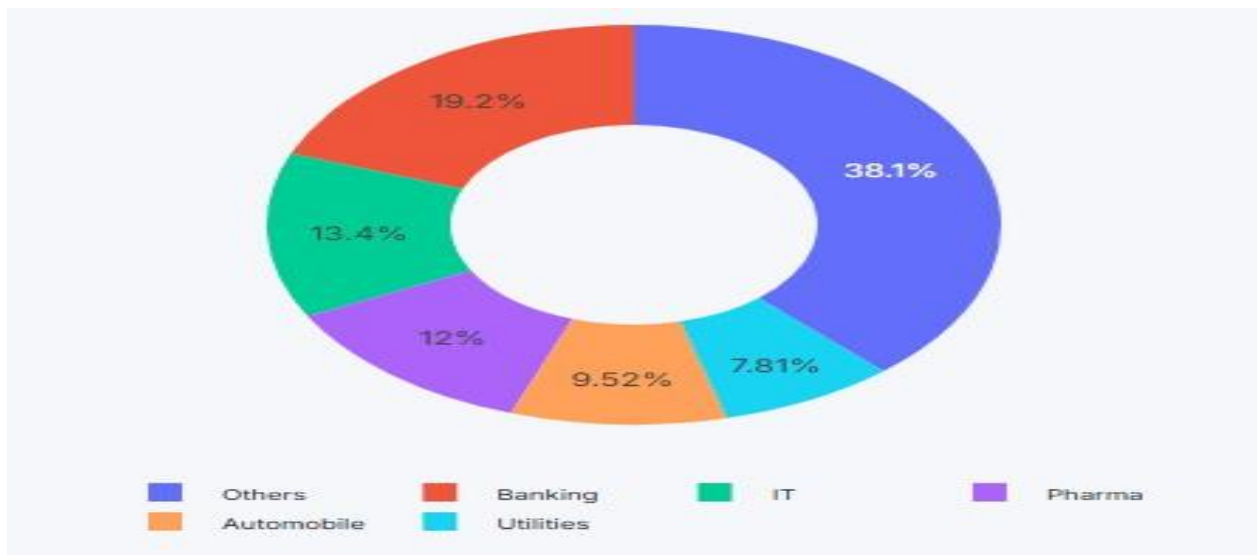
- **File Name:** expense ratio comparison.png | **Page Location:** Risk Analytics (Page 4)
- **Purpose:** Shows the distribution of risk/volatility characteristics across the fund universe.
- **X-Axis / Y-Axis:** Volatility Range % (X) vs Fund Count (Y).
- **Applied Filters:** Aggregated across all active funds.
- **Business Insight:** Highlights that the majority of funds fall within a moderate volatility band (12% to 16%).

7.10 Investor Transaction Distribution



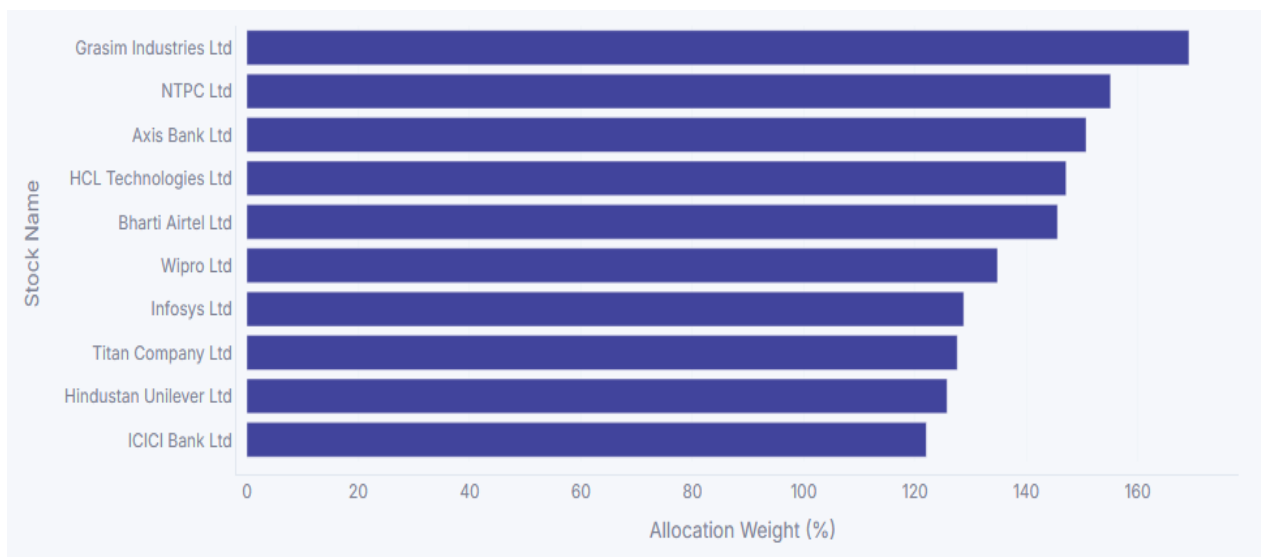
- **File Name:** investor transaction distribution.png | **Page Location:** Investor Analytics (Page 5)
- **Purpose:** Compares the popularity of different investment modes by volume.
- **Metric:** Total transaction volume share in Rs. Crore.
- **Applied Filters:** Aggregated across all demographic categories.
- **Business Insight:** Lumpsum investments dominate total volume, while SIPs represent a growing share of transaction counts.

7.11 Portfolio Sector Allocation



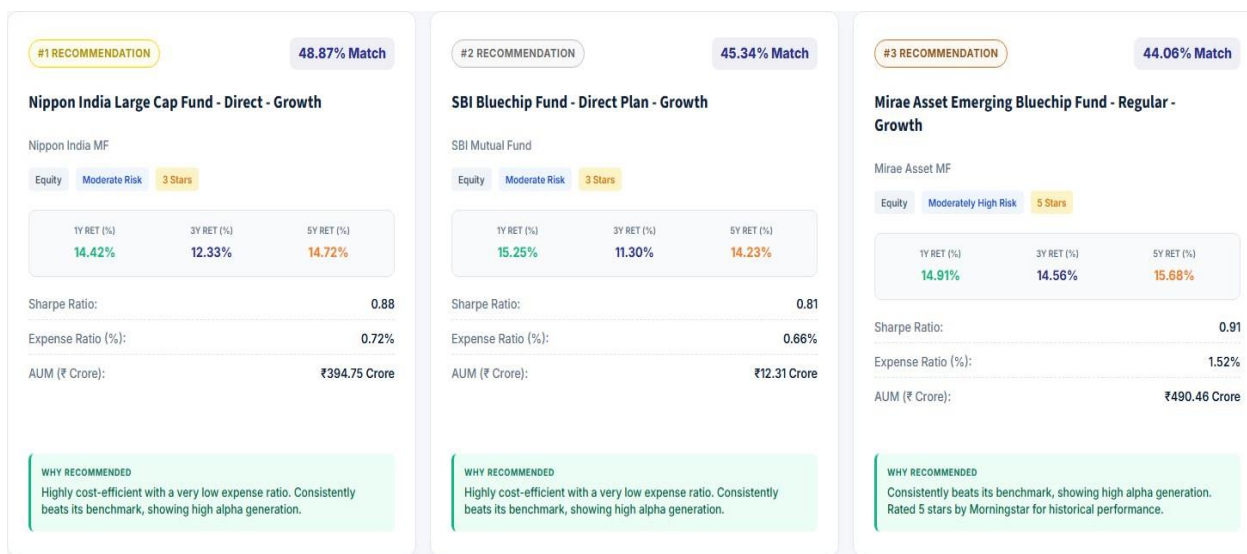
- **File Name:** portfolio sector allocation.png | **Page Location:** Portfolio Holdings (Page 7)
- **Purpose:** Identifies sector weights in the aggregated portfolio.
- **Metric:** Percentage Allocation % (Values) vs Industry Sector (Labels).
- **Applied Filters:** Aggregated across all holdings in the database.
- **Business Insight:** Banking and Financial Services hold the largest allocation share, representing sector concentration.

7.12 Top Holdings Distribution



- **File Name:** top holdings distribution.png | **Page Location:** Portfolio Holdings (Page 7)
- **Purpose:** Highlights the largest individual stock exposures.
- **X-Axis / Y-Axis:** Allocation Weight % (X) vs Stock Name (Y).
- **Applied Filters:** Filtered for the top 10 stocks in descending order.
- **Business Insight:** Highlights exposure to large-cap giants like ICICI Bank, Infosys, and Grasim Industries.

7.13 Fund Recommendation Results



- **File Name:** fund_recommendation_results.png | **Page Location:** Fund Recommender (Page 10)
- **Purpose:** Displays matching mutual funds for a selected investor profile.
- **Format:** Three column cards representing top recommendations.
- **Applied Filters:** Goal: Wealth Creation, Risk: Moderate, Horizon: 3–5 Years.
- **Business Insight:** Recommends *Nippon India Large Cap*, *SBI Bluechip*, and *Mirae Asset Emerging Bluechip* based on score matches.

7.14 Recommendation Table

Score (% Match)	Scheme Name	Fund House	Category	Risk Level	Rating (Stars)	1Y Return (%)	3Y Return (%)	5Y Return (%)	Sharpe Ratio	Expense Ratio (%)	AUM (₹ Crore)
48.87%	Nippon India Large Cap Fund - Direct - Growth	Nippon India MF	Equity	Moderate	3	14.42%	12.33%	14.72%	0.88	0.72%	39,475.00
45.34%	SBI Bluechip Fund - Direct Plan - Growth	SBI Mutual Fund	Equity	Moderate	3	15.25%	11.30%	14.23%	0.81	0.66%	1,231.00
44.06%	Mirae Asset Emerging Bluechip Fund - Regular - Growth	Mirae Asset MF	Equity	Moderately High	5	14.91%	14.56%	15.68%	0.91	1.52%	49,046.00
42.40%	Nippon India ETF Nifty 50 BeES	Nippon India MF	Equity	Moderate	5	10.14%	11.77%	12.31%	0.91	0.89%	20,284.00
41.38%	Kotak Flexicap Fund - Regular - Growth	Kotak Mahindra MF	Equity	Moderately High	5	15.74%	15.65%	13.50%	0.98	1.45%	35,012.00
41.31%	UTI Flexi Cap Fund - Regular - Growth	UTI Mutual Fund	Equity	Moderately High	5	17.43%	15.34%	15.78%	0.96	1.64%	17,912.00
40.41%	Axis Bluechip Fund - Direct - Growth	Axis Mutual Fund	Equity	Moderate	4	13.95%	12.14%	13.66%	0.87	0.75%	15,866.00
38.86%	DSP Top 100 Equity Fund - Regular - Growth	DSP Mutual Fund	Equity	Moderate	5	11.96%	12.82%	12.35%	0.92	1.54%	41,828.00
34.84%	Mirae Asset Large Cap Fund - Regular - Growth	Mirae Asset MF	Equity	Moderate	5	15.12%	14.81%	12.68%	1.06	1.46%	11,361.00
34.50%	ICICI Pru Bluechip Fund - Direct - Growth	ICICI Prudential MF	Equity	Moderate	3	14.12%	14.41%	13.02%	1.03	0.80%	41,553.00

- **File Name:** recommendation_table.png | **Page Location:** Fund Recommender (Page 10)
- **Purpose:** Displays the comprehensive match matrix for candidate funds.
- **Format:** Interactive data grid at bottom.
- **Applied Filters:** Goal: Wealth Creation, Risk: Moderate, Horizon: 3–5 Years.
- **Business Insight:** Allows users to compare match percentages, expense ratios, and ratings side-by-side.

8 Conclusion

8.1 Business Insights

- **Cost vs Alpha Tradeoff:** High expense ratios often erode active management gains. Funds with low fees frequently beat higher-cost peers on a net return basis.
- **Risk-Adjusted Efficiency:** Analyzing Sharpe and Sortino ratios shows that some moderate-risk hybrid funds deliver better risk-adjusted returns than pure equity options during corrections.
- **SIP Stabilization Effect:** Monthly SIP volumes show low volatility, acting as a stable funding source for asset managers during market declines.
- **Demographic Segmentation:** State-wise transaction analysis shows that Maharashtra, Delhi, and Punjab represent major investor hubs, indicating targeted marketing opportunities.

8.2 Challenges Faced

- **Selecting Suitable KPIs:** Determining which visual key performance indicators (KPIs) are most relevant for different pages (such as NAV history vs demographic transactions) without overwhelming the dashboard user.
- **Designing Recommendation Logic:** Constructing a balanced scoring system that combines return metrics, risk profiles, and expense ratios to rank fund recommendations.
- **Benchmark Selection & Comparison:** Determining which benchmark indices (such as NIFTY 50 vs NIFTY Mid-cap 100) are appropriate comparison targets for different mutual fund categories to ensure fair outperformance map-ping.

8.3 Conclusion

- Successfully designed and launched the Mutual Fund Analytics Platform during the internship at BlueStock Fintech.
- The platform centralizes fragmented financial datasets, automates price collection, and provides interactive risk-return visual tools.
- Implementing relational SQLite databases, ETL pipelines, and advanced analytics models delivers significant value to wealth managers.
- The project demonstrates that point-based, data-driven interfaces help demystify mutual fund analytics and assist in rational wealth planning.

8.4 References

- **AMFI India (amfiindia.com):** Public mutual fund master list source and regulatory guidelines.
- **MFAPI (mfapi.in):** Open API endpoint utilized for daily mutual fund NAV historical price extraction.
- **Streamlit Framework Documentation:** Technical reference for building multi-page dashboard applications.
- **Plotly Library Reference:** Charting guide for generating interactive lines, bars, heatmaps, and scatter plots.
- **Modern Portfolio Theory (Markowitz, 1952):** Mathematical foundation for portfolio optimization models.